References Cited

Prior Work

Two key preprints (Zenodo, 2026):

Welz, G. (2026a). *A Vision for AI-Powered Knowledge Engines: A Framework for Systematic Knowledge Discovery and Integration*. Preprint. Zenodo.
https://doi.org/10.5281/zenodo.18463304

Welz, G. (2026b). *The Programming Framework: A Universal Method for Process Analysis*. Preprint. Zenodo.
https://doi.org/10.5281/zenodo.18463442

Supporting prototypes and resources:

Welz, G. (2024–2025). *Genome Logic Modeling Project (GLMP): A Microscope for Biological Processes*. Hugging Face Spaces.
https://huggingface.co/spaces/garywelz/glmp

Welz, G. (2024–2025). *Research Paper Metadata Database*. Hugging Face Spaces.
https://huggingface.co/spaces/garywelz/metadata_database

Welz, G. (2024–2025). *Science Video Database*. Hugging Face Spaces.
https://huggingface.co/spaces/garywelz/sciencevideodb

Related Research

AI-Assisted Research and Knowledge Management

Lewis, P., et al. (2020). Retrieval-augmented generation for knowledge-intensive NLP tasks. *Advances in Neural Information Processing Systems*, 33, 9459-9474.

Ammar, W., et al. (2018). Construction of the Literature Graph in Semantic Scholar. *NAACL-HLT*.

Cachola, I., et al. (2020). TLDR: Extreme Summarization of Scientific Documents. *Findings of EMNLP*.

Färber, M., & Jatowt, A. (2020). Citation Recommendation: Approaches and Datasets. *International Journal on Digital Libraries*, 21(4), 375-405.

Human-Centered AI and Interactive Systems

Amershi, S., et al. (2019). Guidelines for Human-AI Interaction. *CHI 2019*.

Bansal, G., et al. (2021). Updates in Human-AI Teams: Understanding and Addressing the Performance/Compatibility Tradeoff. *AAAI 2021*.

Liao, Q. V., & Varshney, L. R. (2021). Human-Centered Explainable AI (XAI): From Algorithms to User Experiences. *arXiv preprint arXiv:2110.10790*.

Multi-Modal Knowledge Representation

Hogan, A., et al. (2021). Knowledge Graphs. *ACM Computing Surveys*, 54(4), 1-37.

Ji, S., et al. (2021). A Survey on Knowledge Graphs: Representation, Acquisition and Applications. *IEEE Transactions on Neural Networks and Learning Systems*, 33(2), 494-514.

Wang, X., et al. (2021). Cross-Modal Retrieval: A Systematic Review of Methods and Future Directions. *ACM Transactions on Multimedia Computing, Communications, and Applications*, 17(3), 1-34.

Reliability and Trust in AI Systems

Bubeck, S., et al. (2023). Sparks of Artificial General Intelligence: Early Experiments with GPT-4. *arXiv preprint arXiv:2303.12712*.

Lin, S., et al. (2022). TruthfulQA: Measuring How Models Mimic Human Falsehoods. *ACL 2022*.

Zhang, T., et al. (2023). Benchmarking Large Language Models for News Summarization. *arXiv preprint arXiv:2301.13848*.

Process Visualization and Computable Artifacts

Heer, J., & Bostock, M. (2010). Crowdsourcing Graphical Perception: Using Mechanical Turk to Assess Visualization Design. *CHI 2010*.

Munzner, T. (2014). *Visualization Analysis and Design*. CRC Press.

Stasko, J., et al. (2018). Visual Analytics for Data Science. *Foundations and Trends in Human-Computer Interaction*, 11(3-4), 154-307.

Le Novère, N., et al. (2009). The Systems Biology Graphical Notation. *Nature Biotechnology*, 27(8), 735-741.

Larkin, J. H., & Simon, H. A. (1987). Why a diagram is (sometimes) worth ten thousand words. *Cognitive Science*, 11(1), 65-100.

Knowledge Representation and Ontology Engineering

Gruber, T. R. (1993). A translation approach to portable ontology specifications. *Knowledge Acquisition*, 5(2), 199-220.

Noy, N. F., & McGuinness, D. L. (2001). Ontology development 101: A guide to creating your first ontology. *Stanford Knowledge Systems Laboratory Technical Report KSL-01-05*.

Berners-Lee, T., Hendler, J., & Lassila, O. (2001). The semantic web. *Scientific American*, 284(5), 34-43.

Uschold, M., & Gruninger, M. (1996). Ontologies: Principles, methods and applications. *Knowledge Engineering Review*, 11(2), 93-136.

LLM-Based Knowledge Extraction

Zelenko, D., Aone, C., & Richardella, A. (2003). Kernel methods for relation extraction. *Journal of Machine Learning Research*, 3, 1083-1106.

Ahn, D. (2006). The stages of event extraction. *Proceedings of the Workshop on Annotating and Reasoning about Time and Events*, 1-8.

White, J., et al. (2023). A prompt pattern catalog to enhance prompt engineering with ChatGPT. *arXiv preprint arXiv:2302.11382*.

Note on Publications: Two preprints (Welz, 2026a) frame this work: the Knowledge Engine vision and the Programming Framework. The proposed research will generate additional peer-reviewed publications as project deliverables. The existing operational prototypes (cited above) and these preprints serve as the primary evidence of prior work and technical feasibility.